

Predicting Age of Abalone

Daniela Melo, April 30th 2026



Data Overview

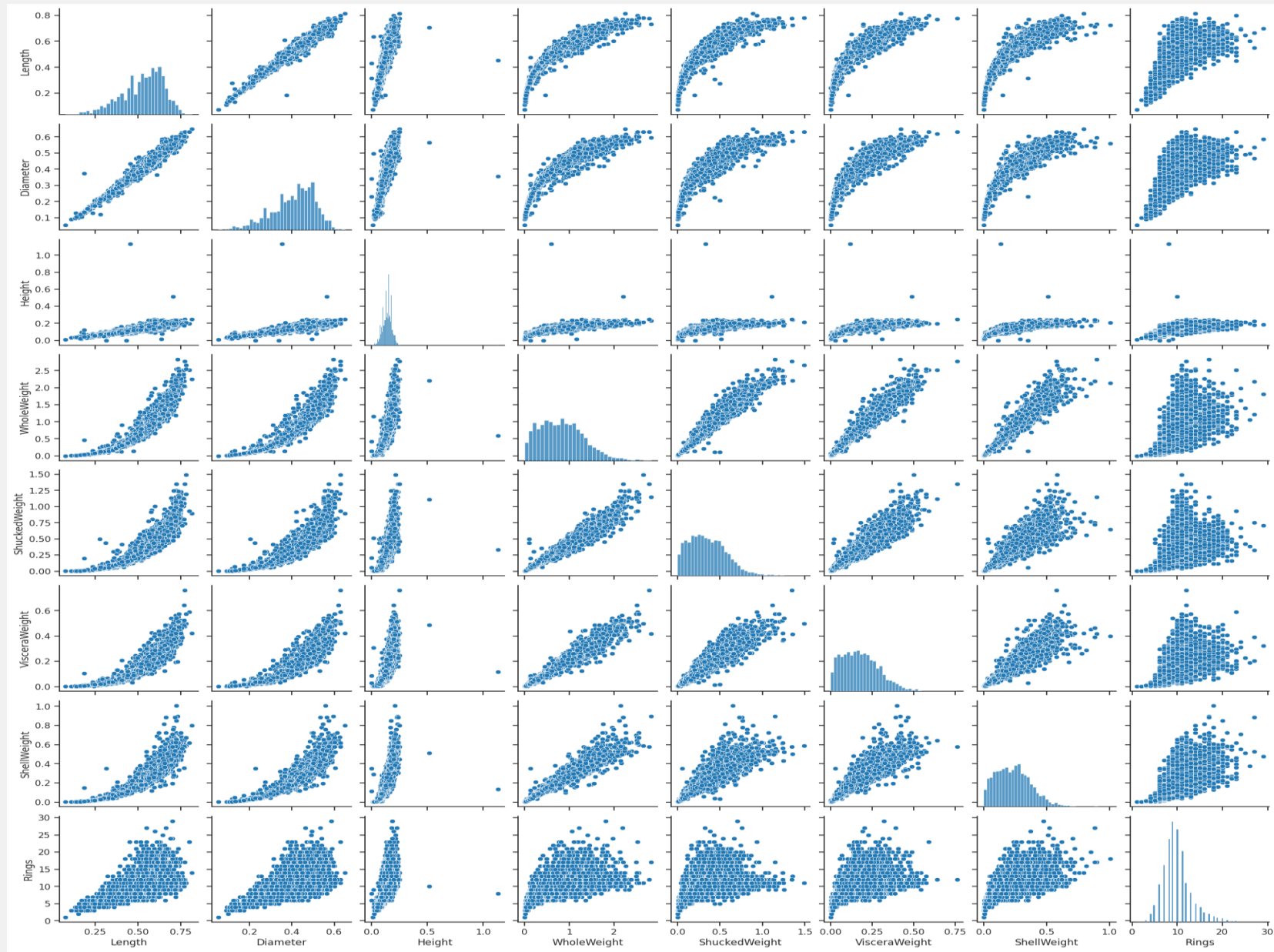
Data from 1994 study by the Department of Primary Industry and Fisheries in Tasmania, Australia.

Available for download from UC Irvine's Machine Learning Repository as a csv.

Consists of measurements of 8 attributes for 4,177 abalone.

Variable Name	Definition	Variable Type
Sex	Abalone sex (Male, Female, and Infant)	Categorical
Length	Longest shell measurement for abalone.	Numeric
Diameter	Shell measured perpendicular to length.	Numeric
Height	Abalone height with meat in shell.	Numeric
WholeWeight	Weight of whole abalone.	Numeric
ShuckedWeight	Weight of meat only, no shell.	Numeric
VisceraWeight	Weight of viscera, removed.	Numeric
ShellWeight	Dried shell weight	Numeric
Rings	Number of age rings identified (+1.5 gives abalone age in years)	Numeric

Variable Relationships

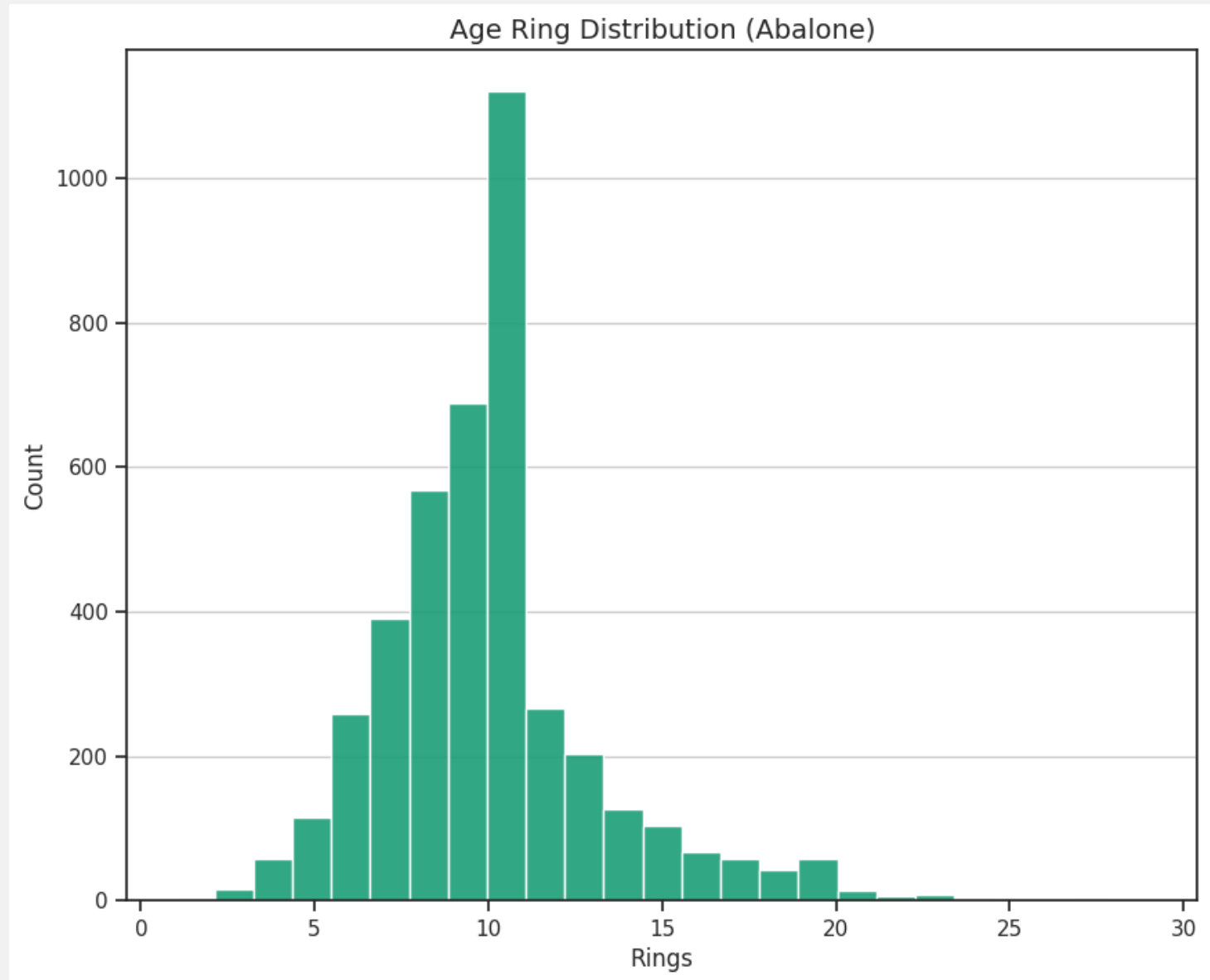


Target Variable

Variable to predict is the number of rings for individual abalone.

Rings can be used to determine age of abalone.

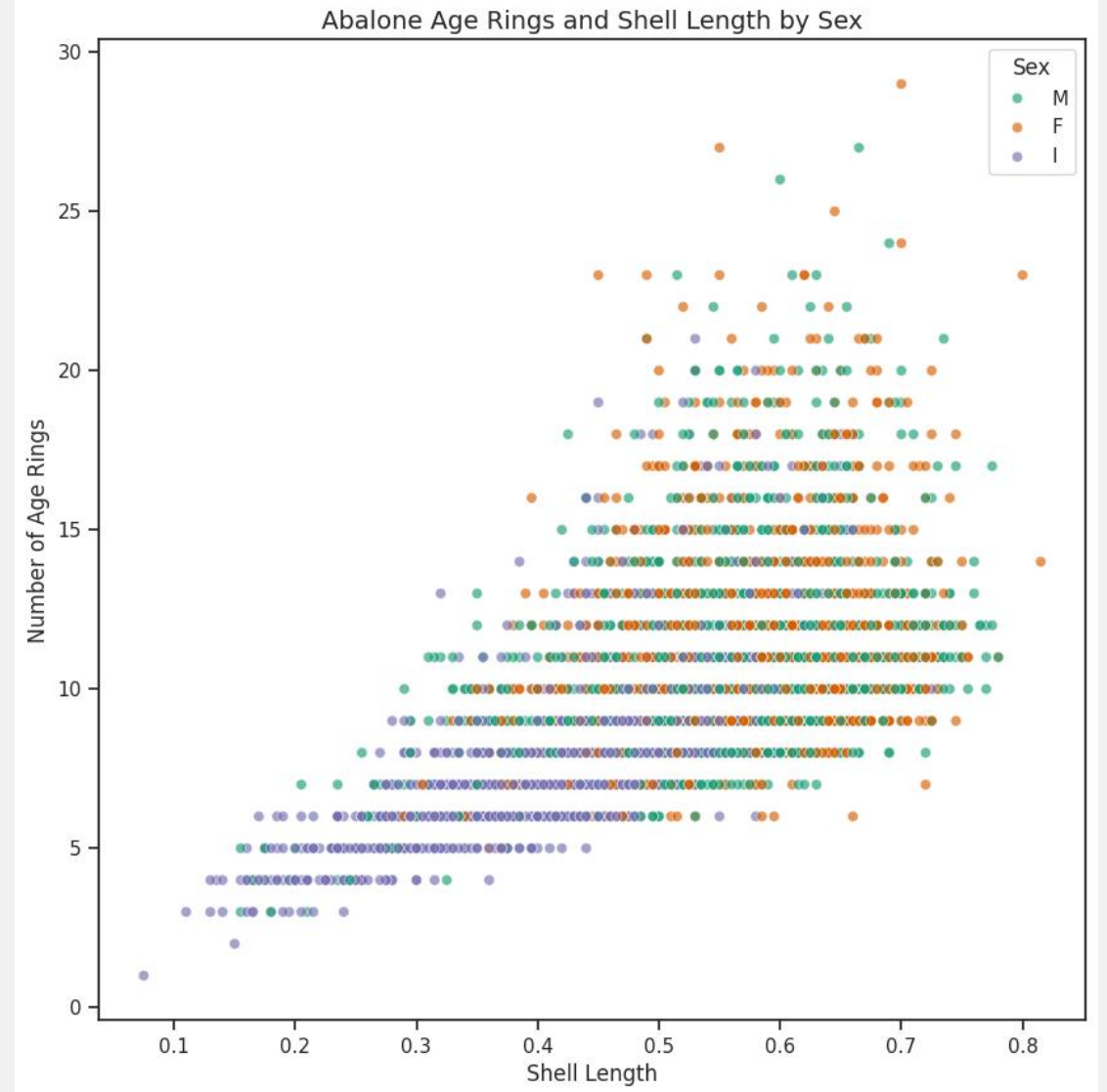
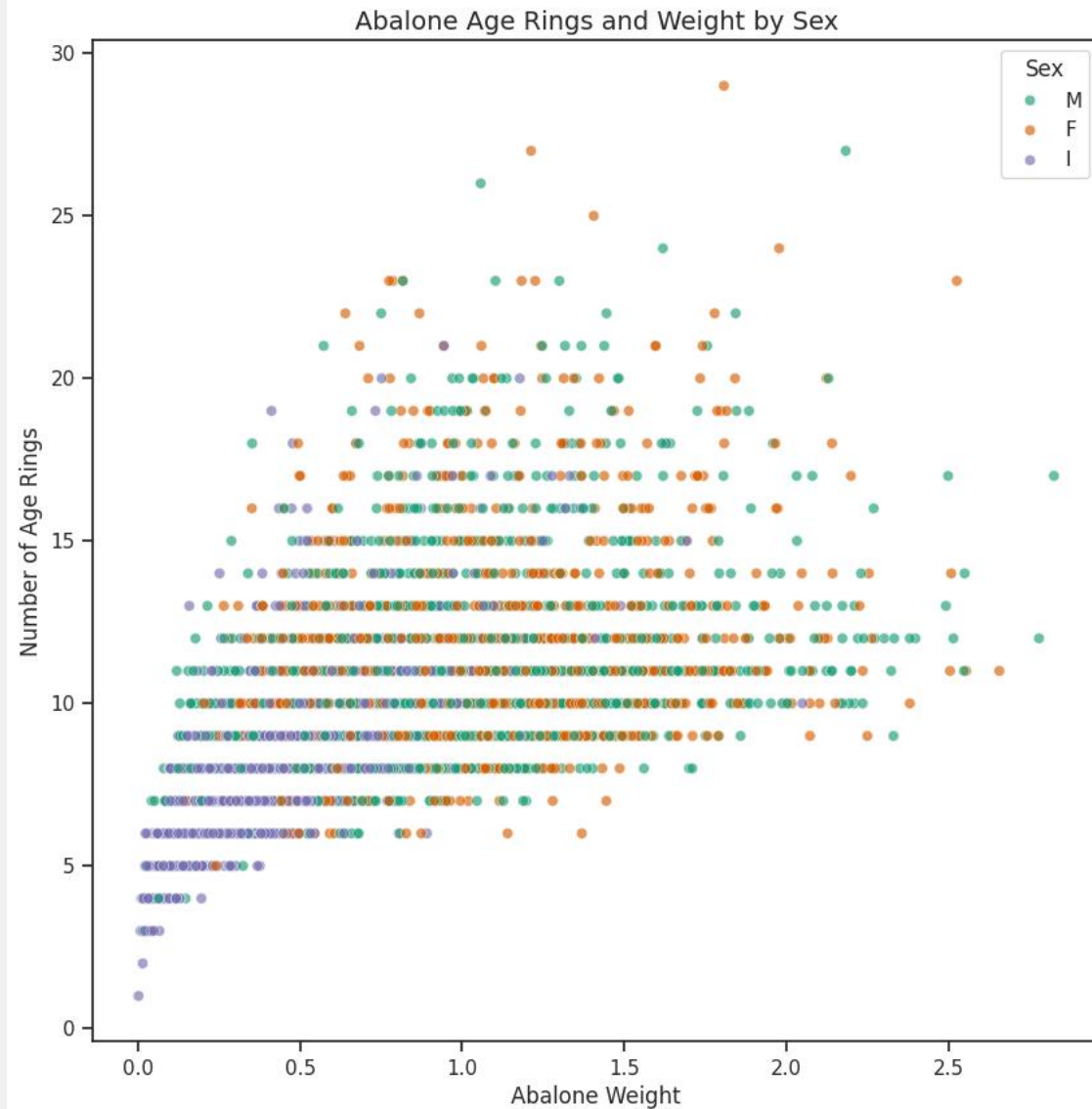
The abalone included in the dataset skew younger.



Predictors

Intent is to use size and weight metrics to predict number of rings.

I chose Shell Length and WholeWeight as my predictive variables



Baseline Model

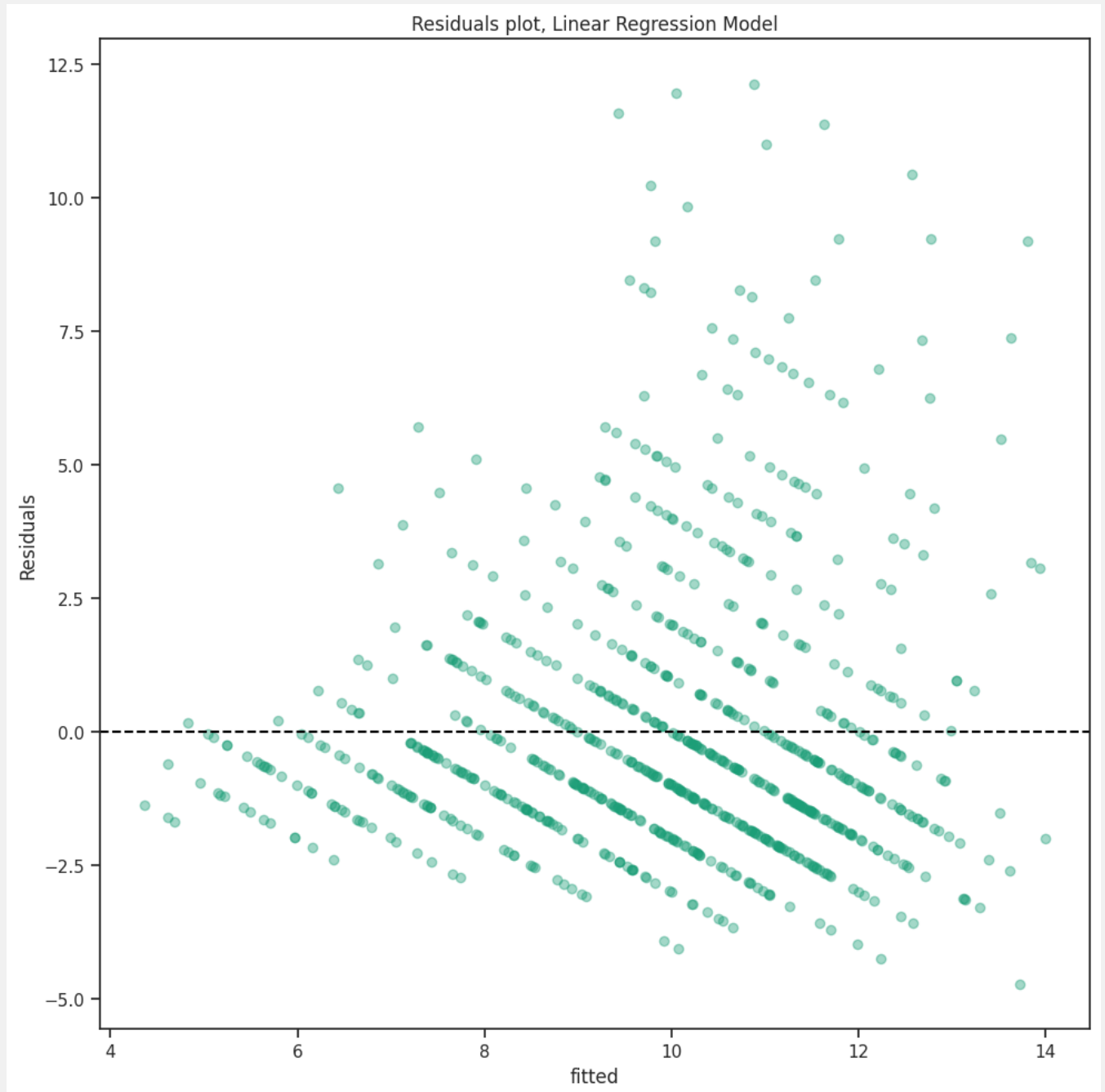
My Baseline Linear Regression did not take Abalone Sex into account.

It predicted 'Rings' as a function of shell height, shell length, and whole weight.

RMSE: 6.87354

R squared: 0.365044

As can be seen from the residuals plot and RMSE/R2, this is not a good model.



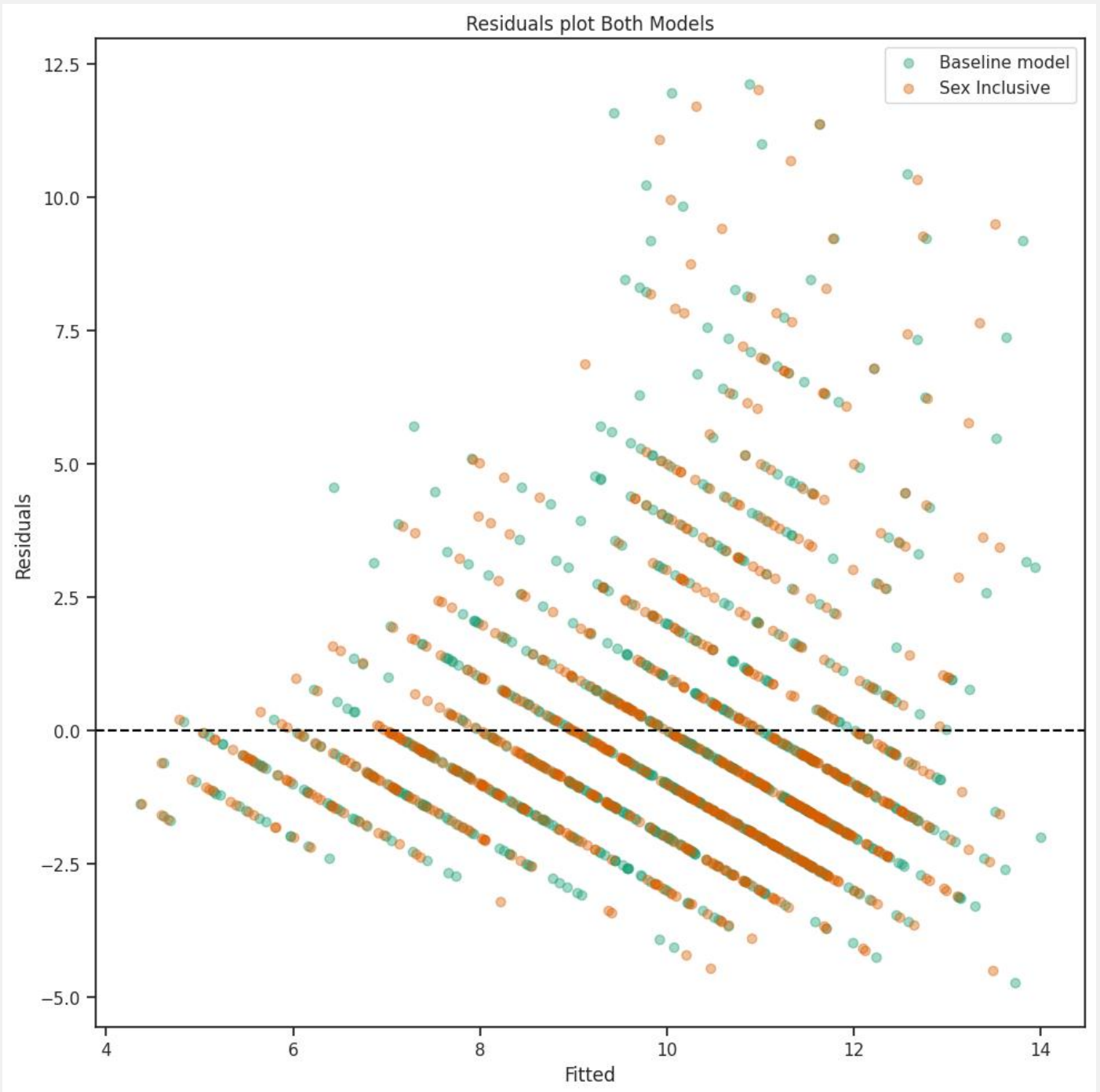
Baseline Model + Sex

Since Sex is a categorical variable, had to hot-encode it into the model.

Updated model fit was slightly better performing, but still not great.

RMSE:6.59456

R squared: 0.390815



Comparing Models

Since residuals were clearly fanned out, I chose to test a couple of different models to see if a transformation may improve model fit.

I also tested a Poisson regression since they tend to be a good fit for count responses, like my response variable ('Rings'), but the best performing model was a 2 degree polynomial model, though it was only slightly better than the Baseline linear model with Sex included.

Model	RMSE	R2
Baseline Linear	6.873537	0.365044
Baseline + Sex	6.594559	0.390815
log	6.871566	0.365226
poly (2)	6.546039	0.395297
Poisson	6.781188	0.373574

Conclusion

The 2nd degree polynomial model is the best fitting model tested, but it still only explains ~ 39.5% of the variation in the data.

This is kind of expected for biological data, and it does indicate that size and sex are actual predictors for age when you account for interaction terms.

Still, it is possible that additional variables related to environment would need to be included in the model to further explain variability.